



Learning Python - Sheet #5

Our first game

Now that we know how to create a program, we can make our first game!

1. Our game

Create a new file (**File > New File**)

Let's type the following instructions, from previous sheet, in the file:

```
import random

MIN = 0
MAX = 100

number = random.randint(MIN, MAX)

print("I'm thinking of a number between %s and %s, can you guess it?" % (MIN, MAX))

while True:
    guess = int(input("What's your guess? "))
    if guess < number:
        print("My number is bigger than %s" % guess)
    elif guess > number:
        print("My number is smaller than %s" % guess)
    else:
        print("Well done! My number is indeed %s" % number)
        break

print("GAME OVER!")
```



Pay attention:

Here are a few things to not in this code:

- Note the first line: **import random**. The keyword **import** tells to python that we want to use a library. The library **random** is part of python's standard library and provides functions to generate and use random numbers.
- We define some variables **MIN** and **MAX**. That's because these numbers are used in two places in the program, and although they are fixed for now, we may want to be able to easily change them in the future.
- Note the infinite loop (**while True**) with the **break** when the player finds the right number



Practice to progress:

You can play a bit with the game, but it lacks some things:

- can you make it count your guesses and tell you how many it took you to find the number?
- can you make it start a new game as soon as you find the number? **Tip:** to make it easier to think about your code, you may want to create a function for a round of the game
- can you make it measure the time it takes you to find the number? **Tip:** you can **import time** and use **time.time()** twice: once at the start and once at the end. The difference between these 2 values will be the number of seconds. It will be a decimal number, you can use **int(end-start)** to make it a whole number or use **round(end-start, 2)** to only keep 2 decimals so that it's more readable.
- can you make the game have multiple levels with different difficulties?



Have fun experimenting:

This game can be really fun. Play with it a lot. What strategy allows you to win with the smallest number of guesses? You think it's just a game? It actually hides something very useful in computer science called **bisection**.

As you play, do you find ideas on how to make this game better? That's the great thing about programming: you can always come up with new ideas to make your program better.

